

Q.4a (20 marks): Evolutionary mechanisms that alter genetic diversity within and between populations

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Q4. (a) Briefly review the ways in which evolutionary mechanisms can alter or structure the genetic diversity within and between populations.

According to the Synthetic Theory,
"Evolution is change in gene frequency"

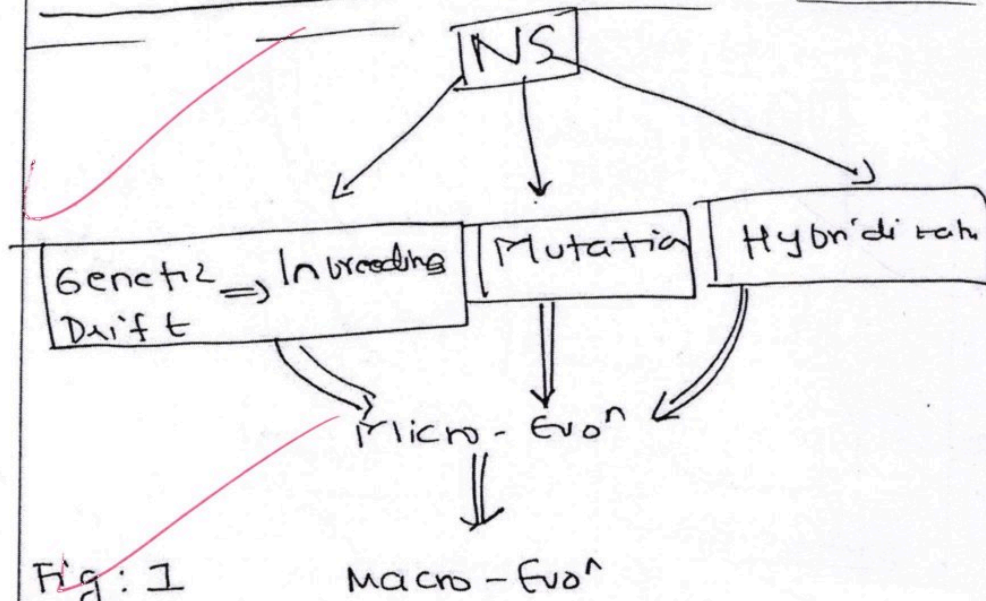


Fig: 1

① Mutation

Sudden random change in the nature of single Gene.

→ Rare; Sudden; Random; recurrent

→ Most Harmful; many neutral and few are helpful.

→ X-ray, γ-ray, ³⁴Chemicals and temp are causes → Mutagenic Agent.

It is source of new-variation

② Change in Chromosome Number L2A™

↓

<u>Euploidy</u>		<u>Aneuploids</u>
$n, 2n, 4n, \text{etc}$	}	$2n \pm 1, \text{etc.}$

In Humans $\Rightarrow$ Euploidy leads to Spontaneous abortion (15% of such cases)

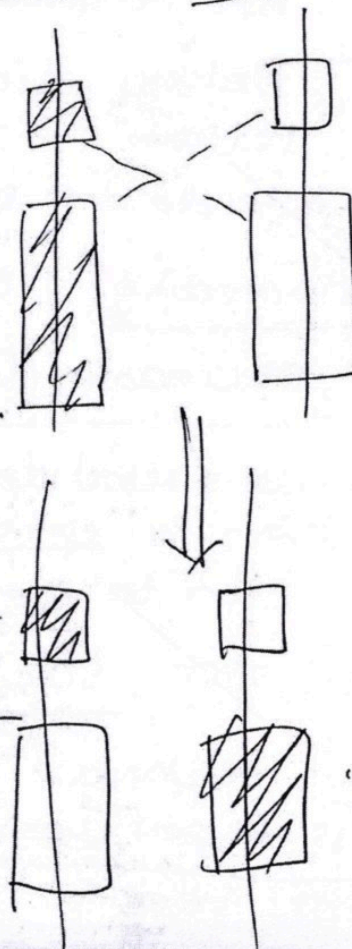
But aneuploidy depends on the specific chromosome and Addition/deletion

③ Genetic Recombination

Fig 2

The order of the genes changes. In humans it is deadly and the effects are the same as change in chromosome number.

Infact: in humans it only leads to expression of the already present variation and not a source of new variation.



"CROSSING OVER"

④ Migration: called gene flow; **L2ATM**
and only valid if permanent mort.
along with mating occurs. Still it
can be understood by hybridization.

⑤ Hybridization: Mating outside a
mendelian population.

→ Increases genotypic variability (GV)

→ Decreases homozygosity

→ Natural selection acts slowly
on it until recessive phenotypes appear.

→ Delays elimination of recessive
alleles as they spread

→ leads to spread of variation

⑥ Inbreeding: Opposite of Hybridization.

→ Decreases (GV) & homozygosity (↑)

→ Considered dominant by Swantham
Thom in speciation

Ex: can be geographical (Sentinel Island)

or even social [caste system]

Ex: Human Genome Project

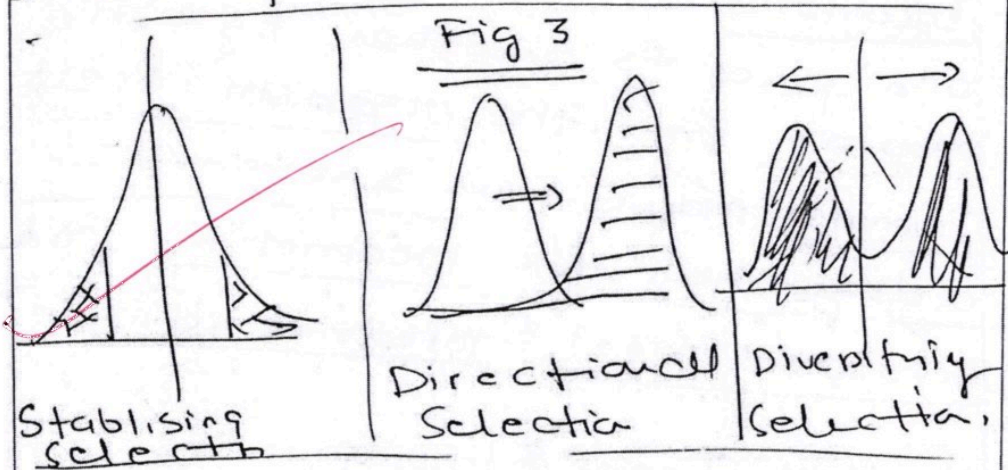
1st Report (2003) said that Africa
has least genotypic variability

hence it is source of all humans.

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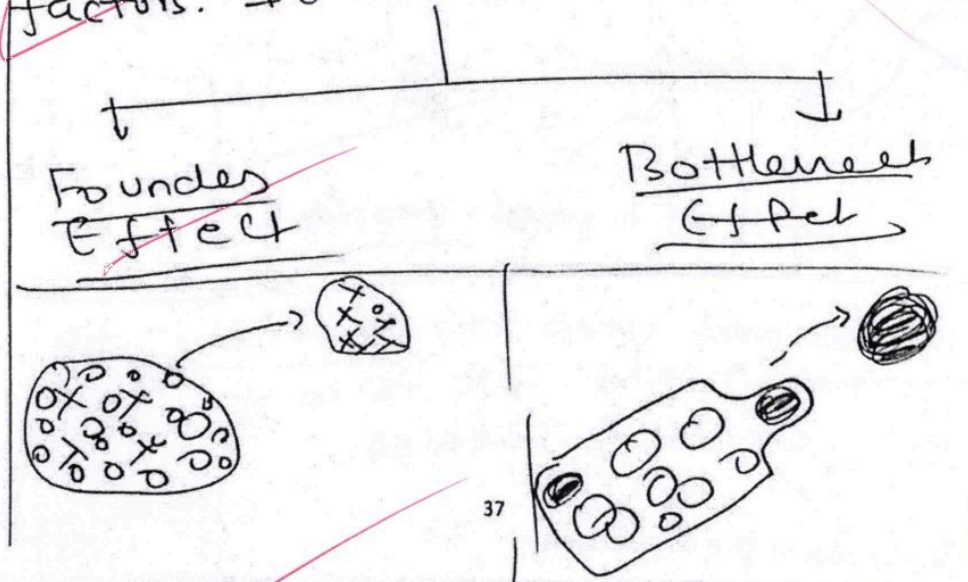
⑦ Natural Selection: It only acts on other factors and is not a source of new variation.

Do not write anything in this margin



⑧ Genetic Drift

Chance & random factor. It is not due to any of the other factors. It is dubbed here.



Thus the mechanisms have had a drastic impact in the history of human evolution.